Wenyu Chiou

AI Research & Engineering | Behavioral Simulation, LLM Evaluation & Agent Systems

Bethlehem, PA · wec324@lehigh.edu · +1 (484) 937-2528 · github.com/WenyuChiou · linkedin.com/in/wenyu-chiou · wenyuchiou.github.io

Ph.D. researcher building human-behavior simulations and testing whether large language models (LLMs) reproduce real people's response patterns and decision-making relationships. Combines survey analysis, statistical modeling, and agent constraints to assess AI behavior and household flood risk.

Open to Summer 2027 internships.

EXPERIENCE

Graduate Research Assistant — Complex Adaptive Water Systems Lab, Lehigh University — Aug 2024 – present

- **Human-grounded LLM evaluation:** Built a Python/Ollama workflow comparing human and Gemma 4:12B response distributions and decision-making pathways. Ten-run median estimates agreed with humans in direction and significance on only 5 of 12 structural equation modeling (SEM) pathways; social-capital response standard deviation was 74-76% lower, exposing lost behavioral variation.
- **Water Agent Governance Framework (WAGF):** Designed a governance layer for human-water simulations to block infeasible or inconsistent agent actions. Checks physical, financial, and behavioral constraints plus rating-reason-action consistency; returns targeted feedback for retries and records audit traces before allowing simulation state updates.
- **FLOODABM:** Built coupled household-agent and catastrophe simulations to assess flood damage, insurance payouts, and household financial burden. Compared with no adaptation, modeled homeowner physical damage fell 5.8%, while losses after insurance fell 45.9%: **property damage alone understates financial protection**. Study under revision.

Independent Projects — Open-Source AI Tools & Education — 2024 – present

- **research-hub-pipeline:** Published a PyPI package with Model Context Protocol (MCP) and command-line interfaces linking scholarly search, Zotero imports, Obsidian notes, and NotebookLM brief verification; replaces disconnected research steps with a reproducible workflow.
- **ai-research-skills:** Built 15 reusable skills across eight research stages, covering literature collection and organization, gap identification, research questions, study planning, and manuscript preparation; 250+ GitHub stars.
- **awesome-agentic-ai-zh:** Built and self-hosted a progressive, hands-on AI-agent learning map: seven stages, 23 practice modules, three languages; 7K+ GitHub stars and 900+ forks.

Research Assistant — National Central University, Surface Hydrology Group — Jan – Jun 2024

- Built and calibrated 2D/3D HydroGeoChem groundwater models; evaluated nature-based watershed solutions.

Summer Research Intern — National Science and Technology Center for Disaster Reduction (NCDR), Taiwan — Summer 2023

- Built Python/MATLAB pipelines comparing climate ensembles for the Intergovernmental Panel on Climate Change Sixth Assessment Report.

Undergraduate Research Intern — Institute of Earth Sciences, Academia Sinica, Taiwan — Jul – Aug 2020

- Contributed stable-isotope measurements to hydrology research at Feitsui Reservoir.

SELECTED RESEARCH OUTPUTS

Yang, Y. C. E., & **Chiou, W.** (2026). Leveraging large language models for agent-based simulation of human-water system interactions. *Water Resources Research, 62*(6), Article e2025WR042111. https://doi.org/10.1029/2025WR042111

Chiou, W., Yang, Y. C. E., Tanaka, T., Jamrussri, S., & Feng, S. (2026). *Household flood adaptation and financial outcomes: A coupled human-flood modeling analysis of homeowners and renters* [Manuscript under revision]. *Water Resources Research*.

Chiou, W., & Yang, Y. C. E. (2026). *Flood adaptation decision variables and pathways across social groups: A comparison between human survey responses and responses generated by large language models* [Manuscript in preparation]. Planned submission: *Progress in Disaster Science*.

Chiou, W., & Chiou, H. (2026). *What makes AI believable? Comparing AI and human responses in mental health risk assessment* [Recorded presentation]. International Society for Data Science and Analytics (ISDSA) Annual Meeting.

Chiou, W., Yang, Y. C. E., Tanaka, T., Jamrussri, S., & Feng, S. (2025). *Modeling long-term household flood adaptation under social heterogeneity: A coupled agent-based modeling framework* [Poster presentation]. American Geophysical Union (AGU) Fall Meeting.

Chiou, W., Yang, E., Sapat, A., Casagrande, D., Tanaka, T., Otsuyama, K., & Yotsui, S. (2025, July 21). *Investigating the decision-making process and causal relationships between flood risk perception and adaptation actions* [Oral presentation]. International Sociohydrology Conference (ISHC), Tokyo, Japan.

Additional conference activity: AGU Fall Meeting (2023), submarine groundwater discharge research (conference abstract).

EDUCATION

Ph.D., Civil & Environmental Engineering, Lehigh University — Aug 2024 – Expected December 2027

M.S., Hydrological and Oceanic Sciences, National Central University — Sep 2021 – Jun 2023

B.S., Earth Sciences, National Central University — Sep 2017 – Jun 2021

SKILLS

Quantitative methods: statistical modeling · psychometrics · confirmatory factor analysis (CFA) · structural equation modeling (SEM) · Bayesian calibration · Monte Carlo · sensitivity and counterfactual analysis · experimental design · behavioral simulation · agent-based modeling (ABM)

Programming & research engineering: Python · R · MATLAB · pytest · GitHub Actions / continuous integration (CI) · Git · PyPI · command-line and REST API development · reproducible pipelines

Machine learning & agent systems: human-grounded large language model (LLM) evaluation · structured generation · prompt/context engineering · multi-agent workflows · Model Context Protocol (MCP) · Codebase Memory MCP · OpenAI / Anthropic APIs · Ollama · LangChain